```
2+3
2*4
2**3
2^3
sin(30)
sin(pi/6)
x=6 #observe environment window
y=c(1,2,3)
sum(y)
max(y)
mean(y)


#import PDF file with code and O/P
#file>compile report>pdf>compile

x*y
z=c(2,3)
p=y*z
log (10)
log(10,10)
log(10,2)

#Vector presentation
y=(1,2,3)# error
y=1,2,3,4#error
y=[1,2,3]#error
y=c(1,2,3)
View(y)
print(y)

#Multiplication
2*3
#Addition
2+3
#Substraction
2-3
#Division 3/2
```

```
#Mixed operations2*4-5+3/2
output:4.5

#power 2^3 30r 2**3
output 8

#Vector power
c(2,3,5,7)^2
output 4,9,25,49

#vector multiplication
c(2,3,5,7)^c(2,3)
output:4,27,25,343

#Error c(2,3,5,7)^c(2,3,4)
#scalar multiplication with vector
c(2,3,4,5,6,7)*3
output:6,9,12,15,18,21

# Vector Vector multiplication
c(2,3,5,7)*c(-2,-3,-5,8)
#output:-4,-9,-25,56

c(2,3,5,7)*c(8,9)
#output:16,27,40,63

c(2,3,5,7)*c(8,9,10)
#Output:Warning message

#Scalar addition
c(2,3,5,7)+6
#output:8,9,11,13

#two vectors addition
c(2,3,5,7)+c(-2,-3,-5,8)
#output:0 0 0 15
c(2,3,5,7)+c(8,8,10)
#Output: answer with Warning message
```

```
c(2,3,5,7)%/% c(2,3)#( Fraction part is discarded)
#output:1 1 2 2

c(2,3,5,7)%%2#( Remainder is considered)
#output: 0 1 1 1

c(2,3,5,7)%%c(2,3)
#output:0 0 1 1

#Maximum
max(1.4,2.6,7.8)
#output:7.8

#Minimum
min(1.4,2.6,7.8)
#output:1.4

#Absolute value
abs(-4)
#output:4
abs(c(-2,-3,-4,-5))
#output:2,3,4,5

#square root
sqrt(9)
#output:3
sqrt(c(2,4,9))
#output:1.141,2,3

#sum
sum(c(4,5,9))
#output:18

#product
prod(c(4,5,9))
output:180

# round
```

```
round(4.29)
#output:4
round(4.52)
#output:5

x1=c(1,2,3)       #Observe in environment window num=number
x=1:8         #int= integer
a=c("XYZ",'PQR')          #chr=character
b=c(TRUE,FALSE,TRUE)        #logi=logic
rm(b)    #removes(clear object from work space) b variable OR in environment convert List to grid >tick for
item> use broom
x2=x1^2
print(x2)
#output:1, 4, 9

#Function
a=function(x){x^2}
a(3)
#output:9

a=function(x,y){x+y}
a(2,3)
#output:5

#matrix
A=matrix(nrow=4,ncol=2,data=c(1,2,3,4,5,6,7,8))#data=1:8
B=matrix(nrow=4,ncol=2,data=c(1,2,3,4,5,6,7,8),byrow=T)
print(A)

#output:A=1 5
#2 6
#3 7
#4 8

#Location
A(2,1)
#Output:error
A[2,1]
#output:2
```

```
A[2,]

mode(A)
#output:numeric

attributes(A)
#output: 4,2

B=matrix(nrow=2,ncol=2,data=c("a","b","c","d"))
mode(B)
#Output: character

# Scalar multiple matrix
C=matrix(nrow=2,ncol=4,data=2)
View(C)

#Unit matrix
D=diag(1,nrow=2,ncol=2)
print(D)

E=matrix(nrow=4,ncol=2,data=1:8,byrow=T)
E[2,]
print(E)

P=matrix(nrow = 2,ncol=2,1:4)
Q=matrix(nrow = 2,ncol=2,c(1,0,2,1))
P+Q
P*Q #entrywise multiplication
P%*%Q #Actual matrix multiplication
t(P)#Transpose matrix
det(P)#determinant
solve(P)#inverse matrix
dim(P)#dimension of matrix
#data: i) data framing  ii) inbuilt data  iii)import data

# Data framing
#The function data. frame() creates data frames, tightly coupled collections of variables which share many of
the properties of matrices and of lists, used as the fundamental data structure by most of R's modeling
```

```
software
Names=c("A","B","C")
Subject=c("Maths","Eng","Sci")
Mathmarks=c(2,4,6)
Engmarks=c(5,7,8)
Scimarks=c(8,9,10)
Marks=data.frame(Names,Subject,Mathmarks,Engmarks,Scimarks)
# another way
Marks=data.frame(Names=c("A","B","C"),
                 Subject=c("Maths","Eng","Sci"),
                 Mathmarks=c(2,4,6),
                 Engmarks=c(5,7,8),
                 Scimarks=c(8,9,10))
print(Marks)
View(Marks)

print(Marks[2,])
#Output: 2nd row

print(Marks[1,"Mathmarks"])
#output:2

#4. Binding the Data in R
#i) rbind: create two data sets with same number of columns/variables
stud_name=c("Rita","Anaya","Smita","Tushar")
LAUC_Marks=c(15,15,20,17)
Roll_No=c(1:4)
MC_Marks=c(15,10,8,20)
Grade=c("A","B","C","D")
D1=data.frame(Stud_name,Roll_No,LAUC_Marks,MC_Marks,Grade)
View(D1)

Stud_name=c("Suraj","Pranay")
LAUC_Marks=c(25,22)
Roll_No=c(5,6)
MC_Marks=c(17,10)
Grade=c("A","B")
D2=data.frame(Stud_name,Roll_No,LAUC_Marks,MC_Marks,Grade)
View(D2)
```

```
#Add number of rows with rbind
D_r=rbind(D1, D2)
View(D_r)

#ii) cbind: Add number of columns with cbind
#create a new data frame with same number of rows/observations

D3=data.frame(PRN=c(111,112,113,114,115,116),
              Dept=c("IT","Comp","E&Tc","IT","E&Tc","Comp"))
#Add number of columns with cbind
D_c = cbind(D_r, D3)
View(D_c)
#OR add new column
D_n=D_r$PRN=c(111,112,113,114,115,116)
#Deletion of first observations
l1=D_c[-1,]
#Deletion of second observations
l2=D_c[-2,]
#Deletion of first variable
l3=D_c[,-1]
#Deletion of second variable
l4=D_c[,-2]
#Deletion of first row and second column
L5 = D_c[-1,-2]

#ii)inbuilt data
data()              # opens list of Data sets in package ‘datasets’
data(mtcars)
View(diamonds)
View(CO2)             #opens in new Tab , also check in help window
View(mtcars)            # opens spreadsheet-style viewer
dim(mtcars)               # dimension: rows x columns
nrow(mtcars)               # number of rows
ncol(mtcars)                # number of columns
colnames(mtcars)              # column names or names(mtcars)
mtcars$am                      # fetch "am" column data
mtcars[,9]
mtcars[,"am"]
```

```
mtcars[2,]
mtcars["Fiat X1-9",]      # gives information of Fiat X1-9 row
head(mtcars)          # default first 6 rows
head(mtcars,10)        # first 10 rows
head(mtcars,2)
mtcars[2,]
tail(mtcars)            # last 6 rows
str(mtcars)              # structure: column names + data types
str(iris)
summary(mtcars)               # statistical summary of each column
summary(iris)
View(iris)
summary(iris)
#check for iris data set
colnames(iris)
mean(iris$Sepal.Length)
mode(iris$Species)
typeof(iris$Species)
mode(iris$Sepal.Length)
typeof(iris$Sepal.Length)
#typeof() is highly precise. It distinguishes whether a number is
#stored as a double-precision floating-point decimal ("double") or
#a whole number ("integer").mode() is broader.
#It simplifies both integers and doubles into a single category: "numeric"
max(iris$Sepal.Length)
min(iris$Species)#error
colSums(mtcars)
attributes(iris$Species)
#Output
#$levels
#[1] "setosa" "versicolor" "virginica"
#$class
#[1] "factor"
View(iris)

# How to read(import)/write(export) csv file
# Create a data frame
student <- data.frame(
  Roll_No = c(101, 102, 103),
```

```
  Name = c("Amit", "Priya", "Rahul"),
  Marks = c(85, 90, 78)
)
# Export to CSV
write_csv(student, "student_data.csv")
#student is existing csv file
#create file in folder with new name student_data
a=write_csv(student, "student_data.csv")
#open a data file in environment window with name a
View(a)
print(a)

#import csv file
b=read_csv("student_data.csv")
#imports file in environment window
#student_data.csv is file in folder, .csv extension is must

#write a command to import/export csv file with name "Marks"
library(readr)
EV <- read_csv("electric_vehicles.csv")
#import file,electric_vehicles is actual name with new name EV
write_csv(EV,"EV1.csv")
#export EV(original) file with downloaded name EV1

#additional for trial on different data sets
c=read.csv("electric_vehicles.csv")#import
write.csv(c,"vehicle data.csv")#export

#Export data from R
#Write data frame into comma separated file
electric_vehicles <- read_csv("electric_vehicles_spec_2025.csv (1) (1).csv")
write.csv(student,"student.csv")
write.csv(Final_Data,"Final_Data.csv")
#Write data frame into space separated text file
write.table(Final_Data,file='Final_Data.txt')

install.packages("openxlsx")
library(openxlsx)
write.xlsx(My.Data,"B")
```